

Oncology Pharmaceutical R&D Industry

Industry Overview

Oncology is the most dominant and fastest-growing sector in pharmaceutical research and development (R&D). Global cancer drug spending reached **\$252 billion in 2024** and is projected to nearly **double to \$441 billion by 2029**, far outpacing other therapeutic areas ([IOVIA, 2024](#)). The United States alone accounts for over **45% of global oncology spending**.

This growth is driven by a robust and expanding drug pipeline, with over **700 oncology drugs in development**. Many target previously untreatable cancers using breakthrough technologies like checkpoint inhibitors, CAR-T therapies, and bispecific antibodies ([EvaluatePharma, 2024](#)). Between 2020 and 2024, the FDA approved **132 new oncology drugs**, with approval counts more than doubling compared to a decade ago ([FDA](#)).

Geographically, the U.S. remains the industry's epicenter, especially in regions like Boston/Cambridge and San Francisco. However, **China now accounts for 29% of global oncology trial starts**, up from just 8% in 2013 ([IOVIA, 2024](#)). While Big Pharma companies dominate oncology revenues, a large portion of early-stage innovation now originates from biotech firms and AI-enabled startups.

Industry Stage

Oncology pharmaceutical R&D is best classified as a **growing industry**, based on firm activity, profitability, investment, and employment data:

- **Firm Entry & Exit:** Cancer biotech startups attracted record venture funding in 2021—**\$13.7 billion globally**—before moderating in subsequent years ([Crunchbase, 2022](#)). Large acquisitions (e.g., Pfizer–Seagen) reflect strong exit opportunities.
- **Profitability:** Drugs like *Keytruda* are projected to exceed **\$30 billion annually**. Oncology now commands the largest share of pharma sales and pipeline assets ([EvaluatePharma, 2024](#)).
- **R&D Investment:** Oncology accounts for over **37% of clinical pipeline investment**, and clinical trials continue to rise—with over **2,000 oncology trials launched in 2023**.

([IQVIA, 2024](#)).

- **Employment:** U.S. pharmaceutical employment grew from 283,000 in 2015 to over **352,000 in 2023**, driven in part by oncology demand ([BLS, 2023](#)).

These indicators—rising firm entry, record spending, expanding trials, and workforce growth—confirm that oncology R&D remains in a dynamic expansion phase.

Data Landscape and Analytical Opportunities

A wide range of credible data sources supports oncology R&D analysis:

Source	Maintained by	Use Cases
ClinicalTrials.gov	NIH	Analyze 87,000+ cancer trials (2000–2021) by phase, design, or sponsor.
FDA Oncology Approvals	FDA	Track oncology drug launches, labels, approval speed.
BLS NAICS 3254	U.S. Dept. of Labor	View wage and employment trends in pharma manufacturing.
Crunchbase News	Crunchbase	Track venture flows to cancer AI startups and biotech exits.
NIH RePORTER	NIH	Explore funded AI-oncology projects (e.g., imaging, modeling).
EvaluatePharma	Evaluate Ltd.	Pipeline forecasts, revenue projections for cancer drugs.
IQVIA 2024 Report	IQVIA	Latest data on trial growth, spending, global pipeline.
OECD Pharma R&D	OECD	Compare country-level pharma R&D intensity and investment.

These sources enable time-series visualizations, international comparisons, and evidence-based assessments of AI's growing role in oncology innovation.

Data Visualizations and Economic Insights

To better understand the structural dynamics and innovation trajectory of the oncology pharmaceutical R&D sector, two data visualizations were developed using credible public and private data sources. These charts focus on trial activity by geography and venture capital flows into oncology startups—two indicators that together reveal the shifting landscape of cancer research and the role of artificial intelligence in shaping its future.

The first figure draws on data from ClinicalTrials.gov and IQVIA to compare global oncology clinical trial volumes across leading countries in 2023. These data were compiled using sponsor-reported trial registries and industry tracking of new trial starts. The visualization offers a clear view of where clinical research is most active and how that activity is geographically evolving:

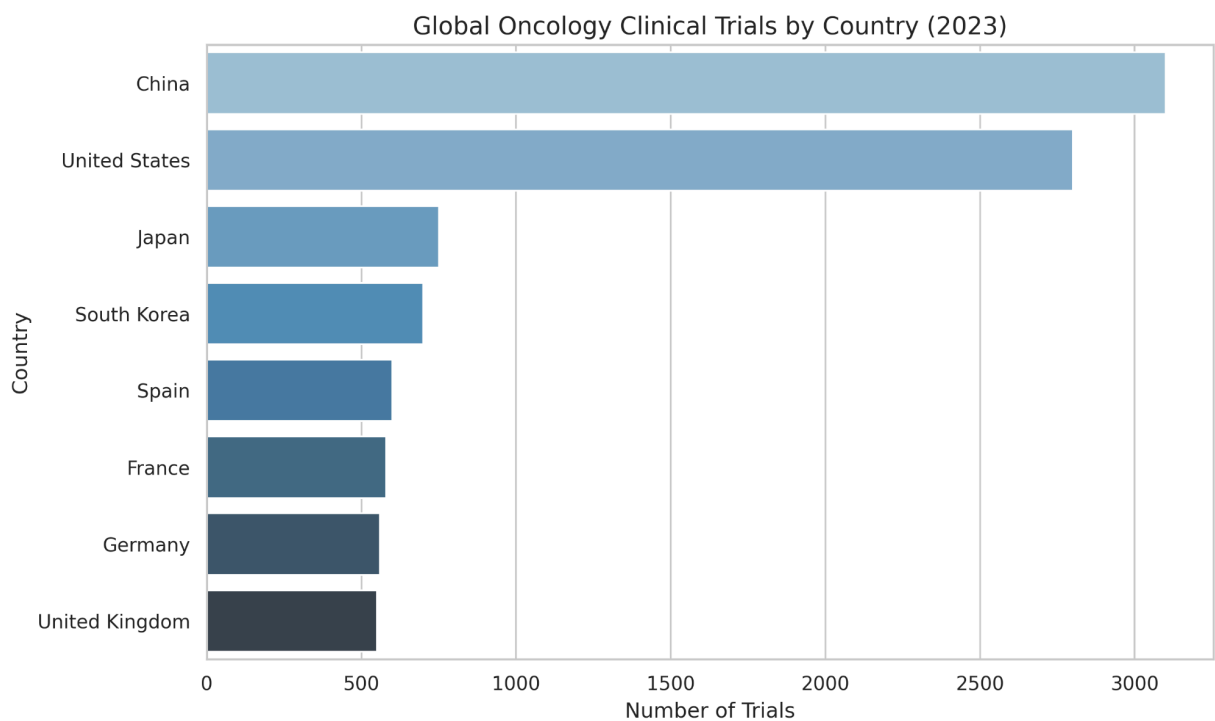


Figure 1. *In 2023, China and the United States led global oncology research, initiating the largest number of clinical trials. China surpassed the U.S. for the first time, reflecting a geographic shift in trial execution. Japan, South Korea, and several European nations followed, each contributing over 500 oncology trials. This global distribution highlights the increasing role of East Asia in early-phase cancer research.*

This visualization reveals a key structural shift in the industry: while the United States remains a global hub for innovation, the center of trial execution is increasingly globalized, with China now leading in oncology trial starts. This may reflect cost dynamics, regulatory flexibility, or expanded domestic demand, and it signals new opportunities—and competitive pressures—for firms and regulators alike.

The second figure illustrates annual venture capital investment in oncology startups between 2015 and 2024, with a focus on AI-enabled companies. Investment totals were derived from Crunchbase and industry deal-tracking reports. The figure distinguishes between total oncology VC and the subset targeting companies that explicitly use artificial intelligence in drug discovery or trial design:

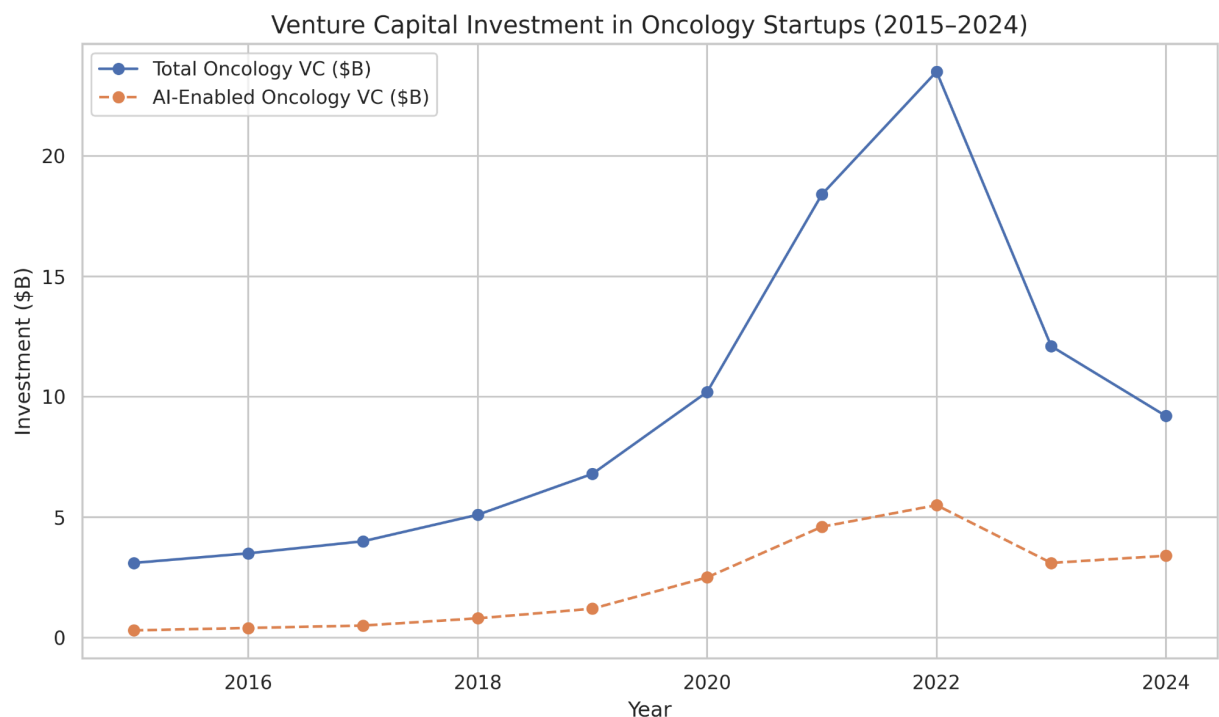


Figure 2. Global venture capital investment in oncology startups peaked in 2021 at \$23.5 billion before declining to \$9.2 billion by 2023. Within this, AI-enabled oncology startups saw rising interest from 2015 to 2021, with funding levels stabilizing around \$3–5 billion in recent years. While overall biotech funding has cooled, AI-focused cancer innovation continues to attract a meaningful share of investment capital.

This investment pattern illustrates how capital flows signal both innovation cycles and investor sentiment. The post-2021 decline aligns with broader biotech deflation, but the resilience of AI-driven oncology ventures suggests that investors view artificial intelligence as a long-term

enabler of drug development efficiency. The continued funding of AI startups—even amid broader market cooling—indicates a structural belief in their productivity advantage.

Together, these visualizations highlight how trial geography and capital flows are reorganizing around technological capabilities, especially AI. They also point to key economic questions about who benefits from this reorganization—and whether AI can expand, rather than concentrate, access to life-saving cancer innovation.

The Impact of Artificial Intelligence on the Industry

Artificial intelligence is poised to transform oncology pharmaceutical R&D, reshaping how clinical trials are conducted, how firms compete, and how workers contribute. This section analyzes the current and emerging impacts of AI across four domains: labor and occupations, firm strategy, risks and harms, and future opportunities. Together, these effects underscore the technology’s dual potential to enhance productivity and widen access—or to exacerbate inequality and concentration—depending on how it is deployed.

1. Impacts on Workers and Occupations

AI is redefining skill requirements and job functions in oncology research. Scientists and trial specialists are now expected to possess computational and data science literacy alongside biomedical expertise. For instance, AstraZeneca launched an AI upskilling initiative in 2024 that certified over 12,000 employees in digital proficiency tiers, enabling lab-based staff to work with algorithmic outputs in drug discovery and trial design (European Pharmaceutical Review, 2024). This reflects a broader workforce transition from manual, repetitive tasks to hybrid roles that integrate scientific judgment with digital tools.

Importantly, AI is functioning as a tool for augmentation rather than replacement. Drug developers use machine learning to screen compound libraries, identify biomarkers, and optimize trial protocols—freeing human experts to focus on hypothesis testing and regulatory strategy. However, this shift is not without labor risk. Routine roles such as data entry and trial monitoring may shrink or evolve, requiring reskilling to prevent displacement. As clinical operations become increasingly automated, firms will need to manage transitional unemployment and invest in workforce development to ensure inclusive adaptation.

2. Impacts on Firms

At the firm level, AI is improving R&D efficiency and altering competitive dynamics. By accelerating early-stage research and streamlining trial design, AI has the potential to cut drug development costs by up to 40% and compress timelines from years to months (McKinsey,

2023). These gains are particularly relevant in oncology, where precision medicine and biomarker-driven trials can benefit from adaptive design and real-time patient stratification. As a result, companies that adopt AI quickly may enjoy a first-mover advantage in both speed to market and portfolio breadth.

The sector has already seen an influx of AI-native startups and novel collaboration models. Large firms like Exelixis and Merck have partnered with AI drug discovery firms such as Insilico Medicine and BenevolentAI, signaling a shift toward decentralized innovation ecosystems. In this new landscape, firms that own or access large training datasets and proprietary algorithms may consolidate market power, while smaller entrants use AI to “punch above their weight.” These trends are reflected in capital flows: as shown in Part (d), VC investment in AI-enabled oncology startups remained resilient even during broader funding downturns, highlighting investor confidence in this disruptive model.

3. Risks and Harms

Despite its promise, AI introduces systemic risks in oncology R&D. One concern is algorithmic bias: if AI systems are trained on underrepresentative datasets—such as trial records that exclude minority populations—they may perpetuate existing disparities in cancer diagnosis and treatment. This is especially troubling in oncology, where biomarker prevalence and treatment efficacy often vary by ethnicity and socioeconomic status (MDPI Bioengineering, 2023). Without corrective measures, AI may optimize for the “average patient” while marginalizing those at the periphery.

There are also labor market and regulatory risks. Automation of entry-level and support roles could lead to workforce dislocation, especially for employees lacking access to reskilling pathways. Additionally, market failures may arise if firms hoard data for competitive advantage, undermining model accuracy and slowing collective progress. Intellectual property concerns over AI-generated drug candidates further complicate the innovation landscape, raising questions about who owns and profits from algorithmically designed molecules. These risks call for updated governance frameworks to ensure that AI advances equity as well as efficiency.

4. Opportunities

At the same time, AI unlocks significant opportunities across product development, firm creation, and workforce evolution. Over 70 AI-discovered drug candidates—including several for cancer—have entered clinical trials as of 2023, demonstrating AI’s capacity to expand the drug pipeline (Nature Biotechnology, 2023). Platforms that simulate molecular interactions or optimize target selection are enabling firms to pursue previously “undruggable” cancers, broadening the scope of innovation. Economic models estimate that AI could generate over \$250 billion in annual operating profit gains for the pharma industry by 2030, with oncology as a major contributor.

AI is also transforming trial logistics. Platforms like Trials.ai and others now help design smarter, faster clinical trials by predicting patient availability, minimizing dropouts, and adjusting protocols dynamically. As shown in Part (d), trial volume is shifting globally, with China now leading in oncology trial starts—reflecting both infrastructure growth and AI-augmented recruitment and site optimization. This geographic redistribution suggests that AI is not only changing how trials are run but also where they happen, potentially redistributing opportunity and influence in the global R&D network.

Finally, AI is catalyzing new career pathways. Demand is rising for bioinformaticians, model engineers, data stewards, and digital trial designers—hybrid roles that blend computational and biomedical expertise. As firms invest in digital capability, a new generation of interdisciplinary workers will shape the next wave of cancer innovation. These labor market shifts may ultimately create a more productive, high-wage workforce—if educational systems and corporate training keep pace.